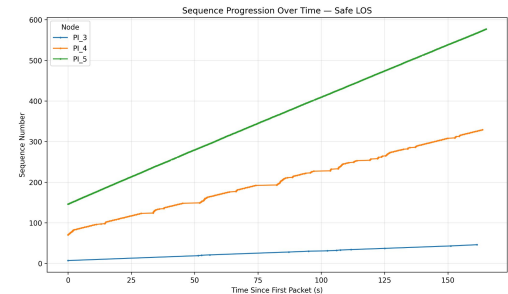
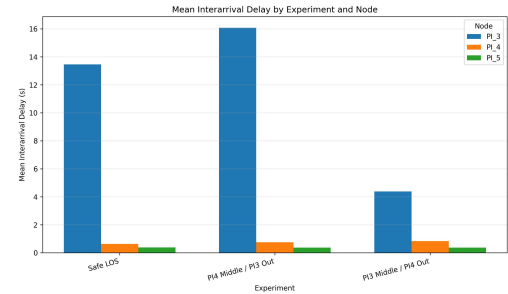
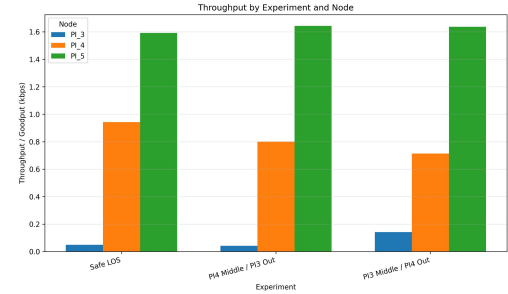


# Drone Garage

4/28/26

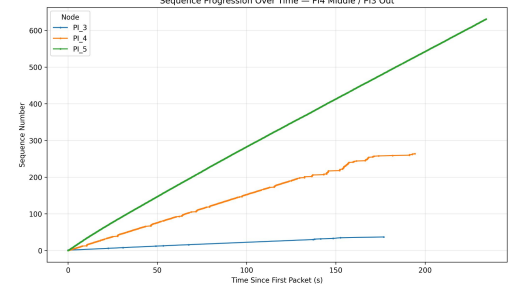
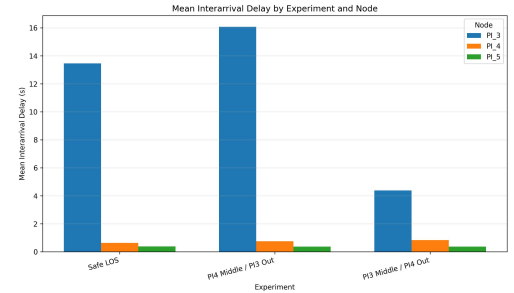
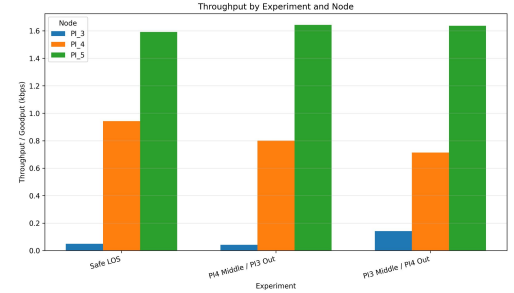
# Safe LoS Analysis

- Safe LoS Setup: With the Pi 5 as the leader, we set both the Pi 3 and the Pi 4 at the same distance from it.
- At this distance
  - The Pi 5 obviously sends the most within the time frame (given its hardware and distance superiority)
  - The Pi 4 sends somewhat steadily, significantly slower than the Pi 5 but also significantly faster than the Pi 3
  - The Pi 3 sends the least, significantly slower than both the Pi 5 and 4



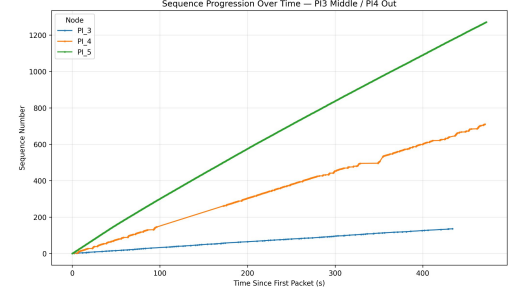
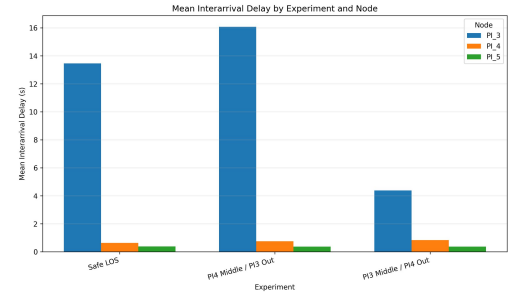
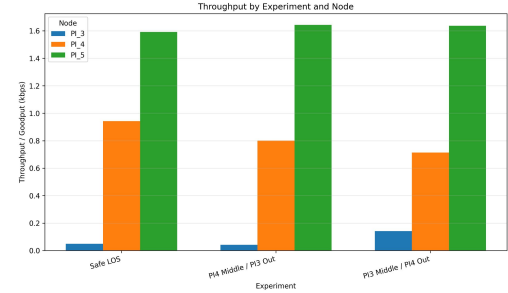
# Pi 4 Middle Analysis

- Pi 4 Middle Setup: With the Pi 5 as the leader, we set both the Pi 4 as the middle node for relaying and the Pi 3 at the “safe” distance from it.
- At this distance
  - The Pi 5 obviously sends the most within the time frame (given its hardware and distance superiority)
  - The Pi 4 sends less than it does in the “Safe LoS” setup
  - The Pi 3 still sends the least and remains the slowest → in fact, it is slower than before



# Pi 3 Middle Analysis (Interesting)

- Pi 3 Middle Setup: With the Pi 5 as the leader, we set the Pi 3 as the middle node for relaying and the Pi 4 at the “safe” distance from it.
- At this distance
  - The Pi 5 obviously sends the most within the time frame (given its hardware and distance superiority)
  - The Pi 4 sends less than it does in the “Pi 4 Middle” setup
  - Interestingly, the Pi 3 is both faster and sends more than it does in the previous two setups
    - Observation: perhaps weaker nodes should be intermediate nodes
    - Observation: Would greater distance between intermediate and end nodes make it so that the messages from the end nodes are coming in so slowly that the weaker, intermediate node can juggle sending those and its own?



# Next Steps

- Mounting Setup on drones
  - 3d print it → find or design → Ask ishan and josh perhaps?
- Visibility-based reorientation
  - At least having the leader determine required coordinate shift → if we're doing this to train a model i suppose we do just a bunch of these?
- Maybe set up 4th Pi to sort of test how an intermediate node can handle two incoming connections?